NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science and Engineering
MID SEMESTER EXAMINATION, Jan-June, 2025
B. Tech (Computer Science and Engineering) 2nd Semester (Sections 1, 2, and 3)
***Course: Computer Organization* (CS24108)**
Full marks:30 Time: 2 hours

Answer *all* questions

Q.no.	Question	Marks	CO	BL
1	a) Discuss the operating steps required to complete the execution of the following two assembly language program instructions (where LOCA denotes a particular location of main memory): Load LOCA, R1 Add R1, R0	05	CO1	Remembering and Application
	b) "Computer systems that contain multiple buses achieve more parallelism."---justify this statement with proper explanation.	04	CO1	Analysis
2	a) "If a 16-bit adder is built by cascading 4-bit carry-lookahead adder blocks, the delays in developing c_{16} and s_{15} are 9 and 10 gate delays, respectively, as compared to 5 and 8 gate delays for the configuration in 16-bit carry-lookahead adder"---analyse how the mentioned gate delays come in both types of designs with all the necessary adder circuit diagrams and mathematical equations. OR a) Solve the following i. Express (- 1.25) in IEEE 32-bit floating points format ii. $(8.844 \times 10^{-3}) - (2.233 \times 10^{-1})$ iii. Compute $(2365)_8 - (1574)_8$ using 7's complement method iv. What will be the equivalent decimal value if the given number uses IEEE 32 bit floating point format 0 01111110 10100000000000000000000	08	CO2	Analysis and Application
	b) Perform the multiplication between the binary forms of +12 and -7 using Booth algorithm.	03	CO2	Application
3	a) Draw all the levels of summand addition tree using 3-2 reducers for the addition of 32 summands during performing fast multiplication. How much full-adder delay will be there in this tree and why? Explain your answer. OR a) i. If a computer system's memory hierarchy were designed with a single, uniform type of memory (e.g., all registers), how would the lack of specialized memory levels (e.g., cache, RAM) impact the system's performance, data retrieval speeds, and overall efficiency in complex computational tasks.	06 03	CO2 CO1	Application and Analysis
	ii. Represent the following decimal numbers in both binary sign/magnitude and twos complement using 16 bits: + 512; - 29.	03		
4	a) Draw the diagram of datapath in a processor and find out the problems that may arise if the multiplexers MuxB and MuxY would not have been there in the datapath. OR (a) Analyze the phases of the instruction cycle (Fetch, Indirect, Interrupt, Execute), highlighting the interdependencies between each stage. Provide a detailed explanation of how time stamps are assigned to each phase, and present a comprehensive diagram that visualizes the flow of operations in the cycle. How do these stages interact in a real-world processor environment?	04	CO2	Remembering and Understanding